Fractions: Concept & Equivalence

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 12

2. B — 18 (27 ÷ 3 = 9, 9 × 2 = 18)

3. 27

4. 20

5. C — 8 (64 ÷ 8 = 8)

6. 2/3

7. 3/4

8. A — 3/5 (18 and 30 share a common factor of 6)

9. 9/12

10. 5/8 (5/8 = 0.625, 3/5 = 0.6)

11. A — 5/6 (5/6 ≈ 0.83, 7/9 ≈ 0.78)

12. 2/5 (2/5 = 0.4, 3/7 ≈ 0.43)

13. 15

14. A and C (21/28 and 6/8 both simplify to 3/4; 9/16 and 5/6 do not)

15. 11/15 (11/15 ≈ 0.73, 7/10 = 0.7)

Standard

1. 1

2. B — 1/3 (7/9 − 4/9 = 3/9 = 1/3)

3. 2 3/5

4. 14/3

5. C — 13/20 (2/5 = 8/20, 1/4 = 5/20, 8/20 + 5/20 = 13/20)

6. 7/12

7. 4 5/6

8. C — 29/8 (3 × 8 + 5 = 29)

9. 11/12

10. 11/18

11. C — 19/20 (7/10 = 14/20, 1/4 = 5/20, 14/20 + 5/20 = 19/20)

12. 11/24

13. 5 2/9

14. A and C (7/12 is the smallest and 3/4 is the largest; order is 7/12, 5/8, 3/4)

15. 3/4, 7/9, 5/6

Standard / Hard

1. 3/4 (LCD 12: 4/12+3/12+2/12=9/12)

2. B — 1/4 (LCD 12: 10/12 − 3/12 − 4/12 = 3/12 = 1/4)

3. 1 5/12 (13/4=39/12, 1 5/6=22/12, 39/12−22/12=17/12)

4. 24 (3/5 of 60=36, 2/3 of 36=24)

5. C — 4 1/2 (3/4 × 6 = 18/4 = 4 1/2)

6. 16 stickers left (2/7×42=12 given to sister, 1/3×42=14 given to friend, 26 given away, 42−26=16)

7. 5/12 left (7 of 12 slices eaten)

8. B — 1 7/12 m (LCD 12: 9/12 + 10/12 = 19/12 = 1 7/12)

9. 1 1/2 cups (2/3 × 9/4 = 18/12)

10. 17/24 (LCD 24: 9/24+8/24)

11. B — 3/10 full (7/10 − 4/10 = 3/10)

12. A, B and C (3/8 × 56 = 21 red, so 35 are not red, which is 5/8 of the total; D and E are incorrect)

13. 13/30 left (2/5+1/6=17/30 spent)

14. 48 km left (3/7×84=36 driven)

15. 1 whole pizza eaten (2/8+6/8=8/8)

Homework — Answers

Easy / Standard

1. 7

2. C — 8 (40 ÷ 5 = 8)

3. 12

4. 15

5. B — 6 (36 ÷ 6 = 6)

6. 3/4

7. 2/3

8. A — 2/3 (12 and 18 share a common factor of 6)

9. 6/9

10. 2/3 is bigger

11. A — 3/4 (3/4 = 6/8, which is greater than 5/8)

12. 4/9 is smaller

13. 6

14. A and B (3/4 and 6/8 both equal 9/12; 5/6 and 7/9 do not)

15. 5/6 is bigger

Standard

1. 19/24

2. B — 13/36 (LCD 36: 28/36 − 15/36 = 13/36)

3. 3 5/6

4. 31/7

5. C — 13/18 (LCD 18: 10/18 + 3/18 = 13/18)

6. 13/24

7. 7 3/4

8. C — 32/9 (3 × 9 + 5 = 32)

9. 11/15

10. 8/15

11. B — 35/36 (LCD 36: 27/36 + 8/36 = 35/36)

12. 11/24

13. 5 3/7

14. A, C and D (order is 7/12, 5/8, 3/4, so 7/12 is smallest, 3/4 is largest, and 5/8 lies between them; B and E are false)

15. 3/4, 7/9, 5/6

Standard / Advanced

1. 3/4 (LCD 12: 3/12+4/12+2/12=9/12)

2. 7/24 (LCD 24: 21/24−8/24−6/24=7/24)

3. 7/12 (as improper: 7/3−7/4 = 28/12−21/12)

4. 18 (2/3 of 45=30, then 3/5 of 30=18)

5. 2 1/2 (5/6 ÷ 1/3 = 5/6 × 3 = 15/6)

6. C — 14 (1/3 × 30 = 10 to brother, 1/5 × 30 = 6 to cousin, 16 given away, 30 − 16 = 14 left)

7. 3/8 left (5/8 eaten)

8. 1 1/24 m

9. B — 1 1/8 cups (3/4 × 3/2 = 9/8 = 1 1/8)

10. 13/20

11. 1/2 full

12. C — 24 (3/7 × 42 = 18 red, so 42 − 18 = 24 are not red)

13. 5/12 left (7/12 spent)

14. A and B (2/5 × 60 = 24 km driven, leaving 36 km; C and D are false)

15. 5/6 eaten